

Mondem Shanwitha Yadav

4th Year Undergraduate

Department of Computer Science Engineering

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology Kanpur	7.9/10
2021	State Board(XII)	Narayana Junior College	98.1%
2019	State Board(X)	Aditya High School	10/10

Projects

• Compiler Design

(Jan'24- Apr'24)

Course project

- Designed a complete **Python** compiler including the phases Lexical Analysis ,Syntax Analysis ,Semantic Analysis ,Intermediate Code Generation and Code Generation.
- **Flex** is used for lexical analysis and **Bison** was used for parsing.
- Using 3ac as the intermediate language the compiler translates python a high-level programming language into x86 a low-level programming language.

• Order Now

(Jun'23- Jul'23)

Self project

GitHub

- A dynamic and feature-rich website has been developed, allowing users to easily browse through our menu with detailed descriptions and enticing visuals and order food items from a variety of specific categories
- The system empowers admins with the ability to edit and manage the menu and food categories, while also offering role-based user registration for enhanced access control and personalized user experiences
- **HTML** and **CSS** and **MySQL** for Database Management, **PHP** for server-side scripting are utilized.

• CSE Bubble

(Mar'23- Apr'23)

Course Project Mentor: Prof. Urbi Chaterjee, Department of Computer Science Engineering

- Built **CSE Bubble** processor having **ISA** similar to **MIPS** with single-cycle instruction fetch, decode and execution
- Designed and implemented the **Arithmetic Logic Unit (ALU)** in top-down approach to develop different modules for **R-type** , **I-type** , **J-type** instructions and the finite state machine for the control signals to execute the CSE Bubble
- Developed MIPS Code for **Bubble Sort** algorithm, translated to machine code following the ISA and executed the processor

• mapGo

(Jan'23- Apr'23)

Course project Mentor: Prof. Indranil Saha, Department of Computer Science Engineering

GitHub

- MapGo is a website developed for the IIT-K junta, which provides the user an all-in one go-to application to navigate through the campus as well as schedule, organise and announce events
- The development team utilized a combination of **Java** for backend development, **HTML** and **CSS** for frontend design, and **MongoDB** for efficient database management in the creation of the Portal
- Various crucial documents, including the Software Requirements Specification Document, Software Design Document, Software Implementation Document, Testing Document, and User Manual are made

Scholastic Achievements

- Secured **All India Rank 2961** in **JEE Advanced 2021** among the 1.4 Lakh shortlisted candidates
- Secured **All India Rank 7635** in common rank list in **JEE Mains 2021** among 9.4 Lakh appeared candidates

Technical Skills

- **Programming:** C, C++
- **Web Development:** HTML, CSS, PHP, SQL
- **Operating Systems:** Windows
- **Hardware Description Language:** Verilog
- **Other Skills:** Bash Scripting, MIPS Assembly Language, OOPS in C++

Relevant Courses

Data Structure and Algorithms Linear algebra and Differential Equations Introduction to Probability Compiler Design Operating Systems	Discrete Mathematics Introduction to Electronics Introduction to Logic Computer Networks Introduction to Machine Learning	Fundamentals of Computing Software Development and Operations Computer Organisation Parallel Computing Theory of Computation
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